IGCSE 0607 Practice Questions — E2.6 — Combined Questions & Markscheme

Inequalities

Total Questions: 51 — **Total Marks:** 277 Marks

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- **E2.6** Inequalities
  → **Paper 2 (Non-Calculator)** *(29 Qs)* · **Paper 4 (Calculator)** *(22 Qs)*

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1 Solve the inequality.

$3x - 5 < 7x + 3$

..... **[2]**

2 **(a)** Solve the inequality.

$$3(x-2) > 5x + 4$$

..... **[2]**

(b) Write down the largest integer that satisfies the inequality in part (a).

..... **[1]**

3 Solve the inequality.

$4(3x - 1) - 5x \geq 2(x + 8)$

..... **[3]**

4 **(a)** Solve the inequality

$$3(x-4) < 5x - 2$$

..... **[2]**

(b) Write down the smallest integer value of x that satisfies this inequality.

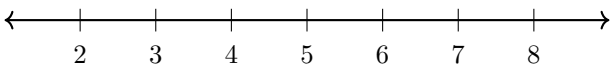
..... **[1]**

5 **(a)** Solve the inequality.

$$3(2x - 1) > 4x + 7$$

..... **[2]**

(b) On the number line, represent the solution to the inequality in part (a).



[1]

(c) Write down the smallest integer value of x that satisfies this inequality.

..... **[1]**

6 **(a)** Solve the inequality.

$$-5 < 3x + 1 \leq 13$$

..... **[2]**

(b) Write down all the integer values of x that satisfy this inequality.

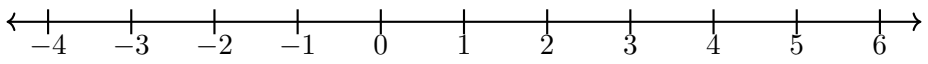
..... **[2]**

7 **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

(b) Show your solution to part (a) on the number line.



[2]

8 **(a)** Solve the inequality.

$4x - 3 \leq 17$

..... **[2]**

(b) Write down the largest integer x that satisfies the inequality.

$3(x + 2) < 20$

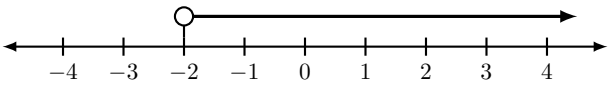
$x =$ ..... **[2]**

9 **(a)** Solve the inequality.

$$4x - 3 < 2x + 9$$

..... **[2]**

(b) Write down the inequality shown on the number line.



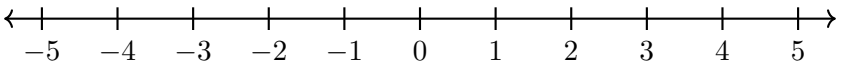
..... **[2]**

10 **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[3]**

(b) Represent your answer to part (a) on the number line provided.



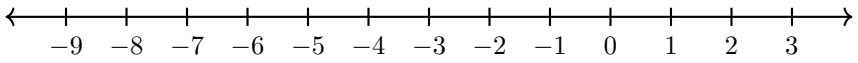
[2]

11 **(a)** Solve the inequality.

$$3(x-2) \leq 5x+8$$

..... **[3]**

(b) Represent the solution to part (a) on the number line.



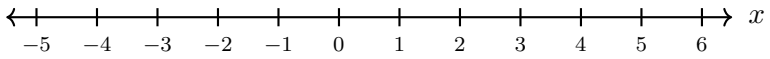
[2]

12 **(a)** Solve the inequality

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

(b) Represent your solution to part (a) on the number line.



[2]

(c) Write down the smallest integer value of x that satisfies the inequality $-5 < 2x - 1 \leq 7$.

..... **[1]**

13 **(a)** Solve the inequality.

$3x - 2 > 5x + 7$

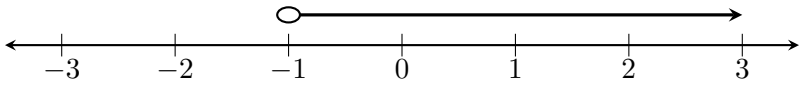
..... **[2]**

(b) Write down all the integer values of n that satisfy the inequality.

$-5 < 2n - 1 \leq 3$

..... **[3]**

(c) An inequality is shown on the number line above.
Write down the inequality.



..... **[1]**

14 **(a)** Solve the inequality.

$$4x - 3 \leq 7x + 12$$

..... **[2]**

(b) Solve the inequality.

$$-8 < 2x - 1 \leq 7$$

..... **[2]**

(c) Write down the largest integer and the smallest integer that satisfy the inequality in part (b).

Largest = ... , Smallest = ... **[2]**

15 **(a)** Solve the inequality.

$2 - 3x < 11$

..... **[2]**

(b) Solve the inequality.

$4(2x - 1) \leq 5x + 17$

..... **[3]**

(c) n is an integer such that $2 - 3n < 11$ and $4(2n - 1) \leq 5n + 17$.
Write down all the possible values of n.

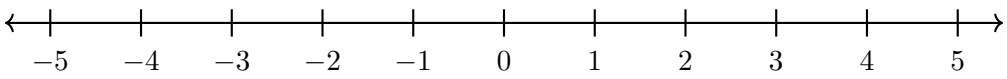
..... **[2]**

16 **(a)** Solve the inequality.

$3x - 5 \leq 7x + 3$

..... **[2]**

(b) On the number line, show the solution to the inequality in part (a).



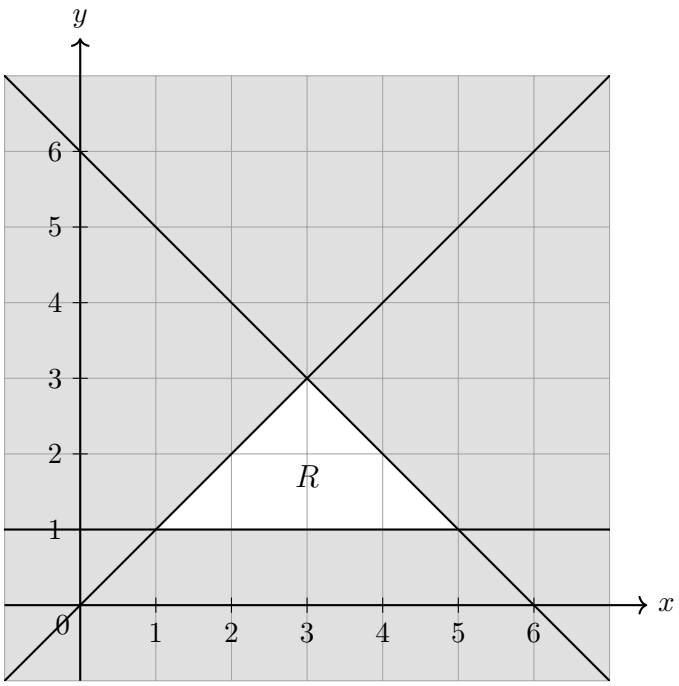
[2]

(c) List all the integers that satisfy the inequality.

$-2 < 3x - 5 \leq 4$

..... **[3]**

17 The unshaded region R in the diagram is defined by three inequalities.



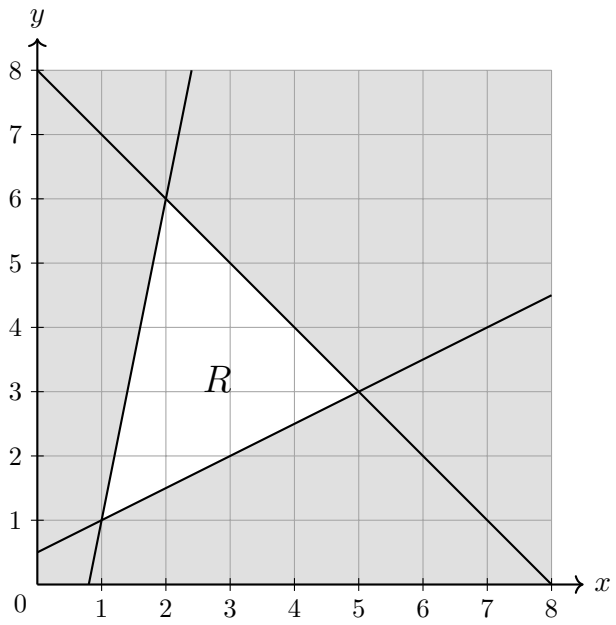
Write down these three inequalities.

.....

.....

..... **[3]**

18 The diagram shows a region R, which is unshaded.
The unwanted regions are shaded.



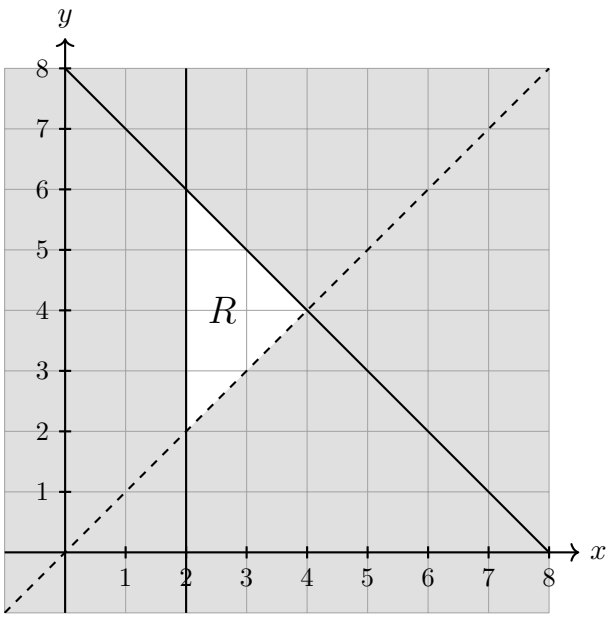
Write down the three inequalities that define region R.

.....

.....

..... **[3]**

19 The diagram shows a region R defined by three inequalities.
The unwanted regions are shaded.

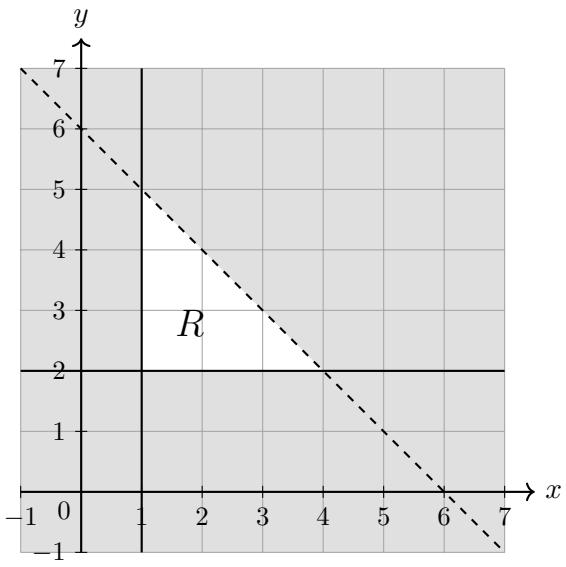


Write down the three inequalities that define the unshaded region R.

.....

..... **[3]**

20 The unshaded region R is defined by three inequalities.
The boundaries of the region are shown on the coordinate grid.



Write down the three inequalities that define the unshaded region R.

.....

.....

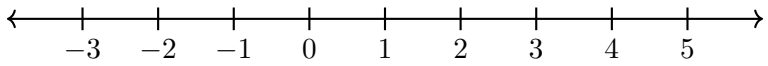
..... **[3]**

21 **(a)** Solve the inequality.

$$-4 < 3x + 2 \leq 14$$

..... **[2]**

(b) Represent your answer to part (a) on the number line.



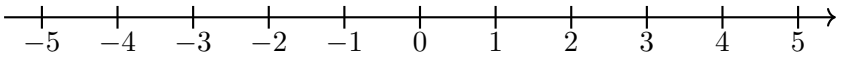
[2]

22 **(a)** Solve the inequality.

$$-7 < 2x - 1 \leq 5$$

..... **[3]**

(b) Represent your answer to part (a) on the number line.



[2]

23 **(a)** Solve the inequality.

$$-5 \leq 2x - 1 < 6$$

(a) Solve the inequality.

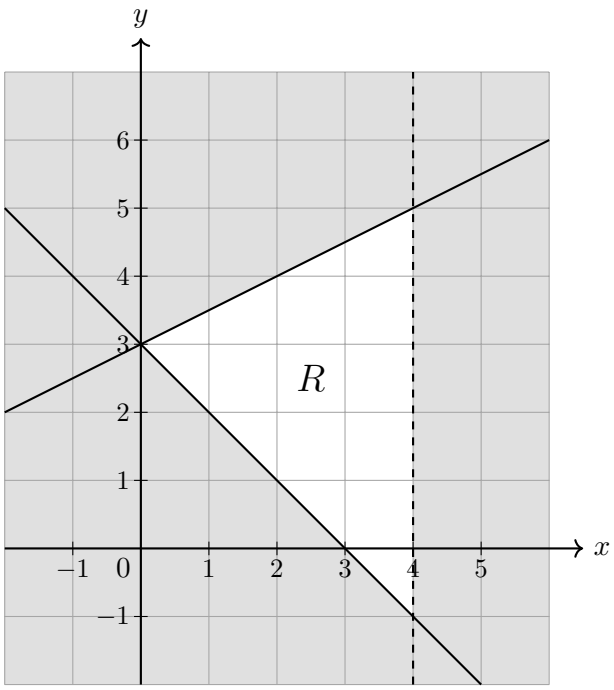
$$-5 \leq 2x - 1 < 6$$

..... **[3]**

(b) Write down all the integer values of x that satisfy this inequality.

..... **[2]**

24 The unshaded region R is defined by three inequalities.



(a) Write down these three inequalities.

.....

.....

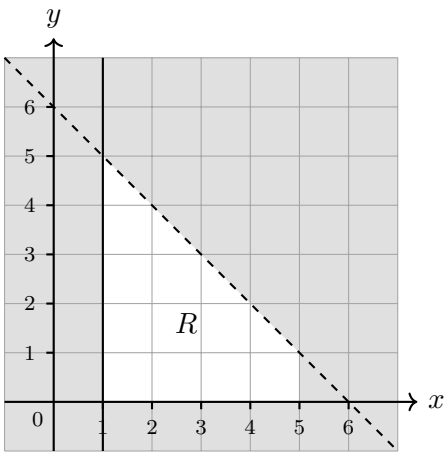
..... **[3]**

(b) The point (a, b), where a and b are integers, lies in the region R.
It is given that $a = 3$.

Find all possible values of b.

..... **[3]**

25 The unshaded region R in the grid is defined by three inequalities.
One of these inequalities is $x \geq 1$.



(a) Write down the other two inequalities that define region R.

.....

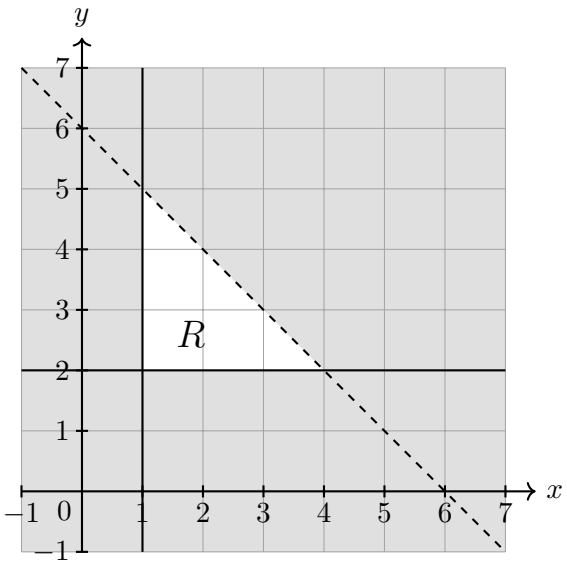
.....

..... **[4]**

(b) Find the maximum value of $2x + 3y$ for points with integer coordinates (x, y) inside region R.

..... **[2]**

26 The diagram shows a region R, which is unshaded.
The region is bounded by three lines.
One of the lines is broken.



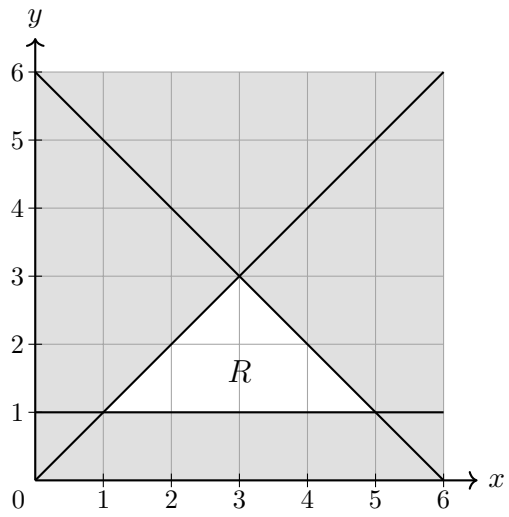
Write down the three inequalities that define the region R.

.....

.....

..... **[3]**

27 The diagram shows a grid with three straight lines and an unshaded region labeled R.



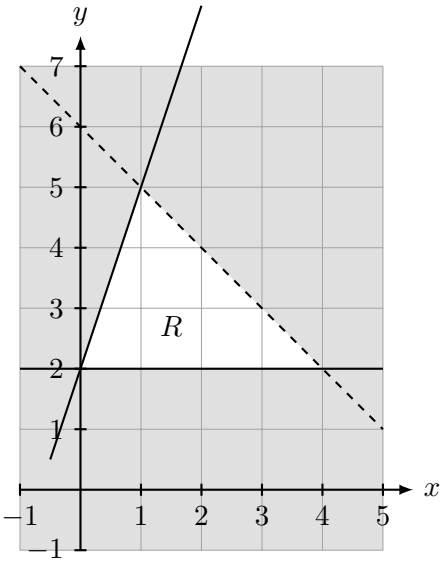
(a) Write down the three inequalities that define the unshaded region R.

[3]

(b) Find the maximum value of $3x + 2y$ for integers x and y that lie in the region R.

..... **[2]**

28 A region R is defined by three inequalities.



The region R, unshaded, is defined by three inequalities.
Write down these three inequalities.

.....

.....

..... **[6]**

29 **(a)** Solve the inequality.

$$\frac{3x-2}{5} - \frac{2x-5}{3} \leq 2$$

..... **[4]**

(b) A region R is defined by the three inequalities $y > x$, $y < 4$, and $x + y > 2$.
Write down all the points in region R that have integer coordinates.

..... **[3]**

30 **(a)** Solve the inequality.

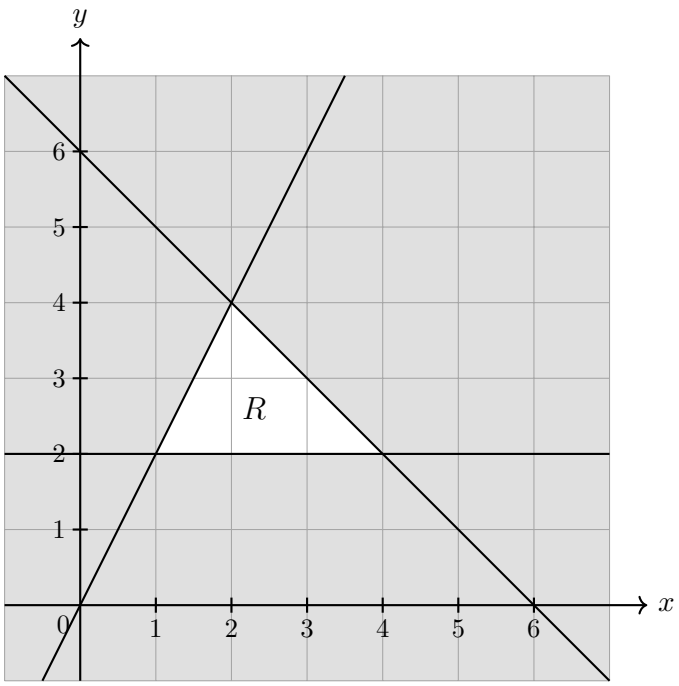
$$5x - 3 \leq 2x + 12$$

..... **[2]**

(b) x is a positive integer.
Write down all the possible values of x.

..... **[2]**

31 The diagram shows a region R unshaded.



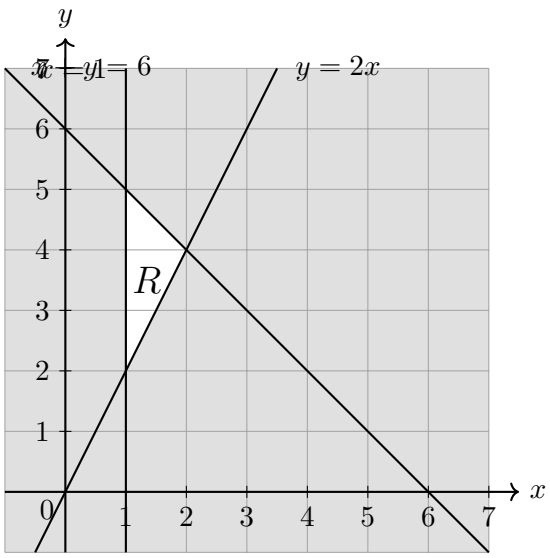
Write down the three inequalities that define the region R.

.....

.....

..... **[3]**

32 The diagram shows three straight lines: $x = 1$, $y = 2x$, and $x + y = 6$.
The unshaded region R is bounded by these lines.



Write down the three inequalities that define the unshaded region R.

.....

.....

..... **[3]**

33 **(a)** Solve the inequality.

$$-7 < 3x + 2 \leq 11$$

..... **[3]**

(b) Write down the largest prime number that satisfies this inequality.

..... **[1]**

34 **(a)** Solve the double inequality.

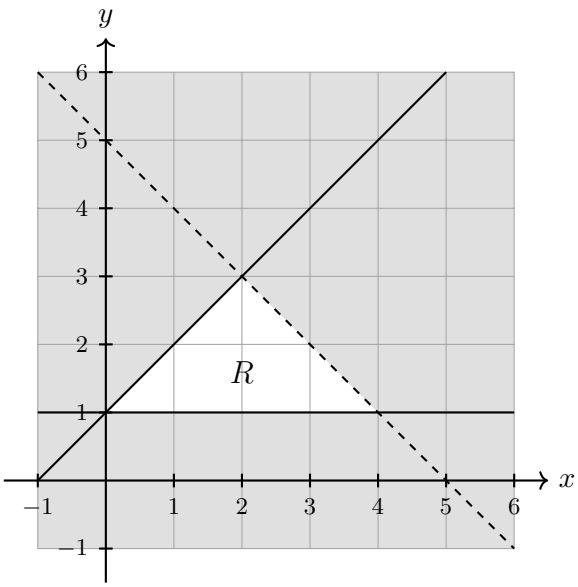
$$-7 \leq 3x - 1 < 11$$

..... **[3]**

(b) Write down all the integer values of x that satisfy the inequality in part (a).

..... **[2]**

35 The diagram shows a region R which is unshaded.
The unwanted regions are shaded.



(a) Write down the three inequalities that define region R.

.....

.....

..... **[4]**

(b) Find the point (x, y), with integer coordinates, inside region R which maximizes the value of $2y - x$.

((..... , ...)) **[2]**

36 **(a)** Solve the inequality.

$$-5 < 2x + 1 \leq 11$$

..... **[3]**

(b) Write down all the integers that satisfy the inequality.

$$-2.5 \leq y < 1$$

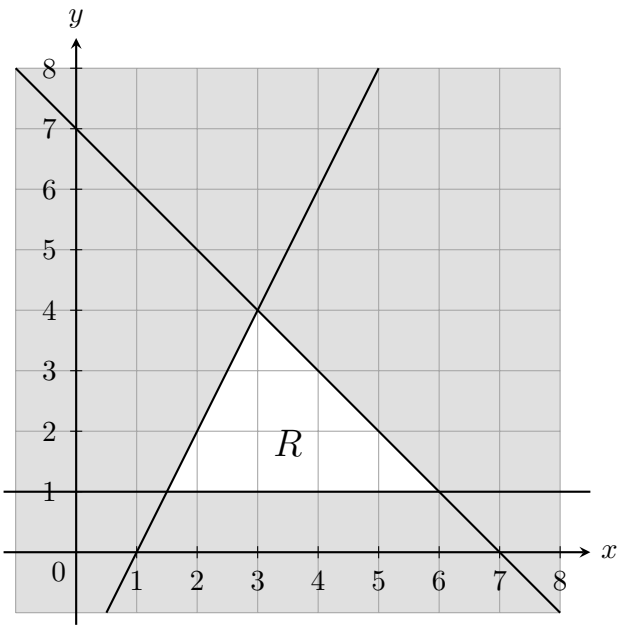
..... **[2]**

(c) Use your graphic display calculator to solve the inequality.

$$2x^2 - 7x + 3 > 0$$

..... **[2]**

37 The diagram shows a grid with three straight lines.



(a) The unshaded region R is defined by three inequalities.
The boundaries are included in the region R.
Write down the three inequalities.

.....

.....

..... **[5]**

(b) A point (x, y) lies in the region R.
Both x and y are integers.
Find the maximum value of $2x + y$.

..... **[2]**

38 **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

(b) Write down all the integers that satisfy the inequality in part (a).

..... **[2]**

(c) A rectangular garden has length $(3x + 2)$ metres and width x metres.
The perimeter of the garden is strictly less than 40 metres.

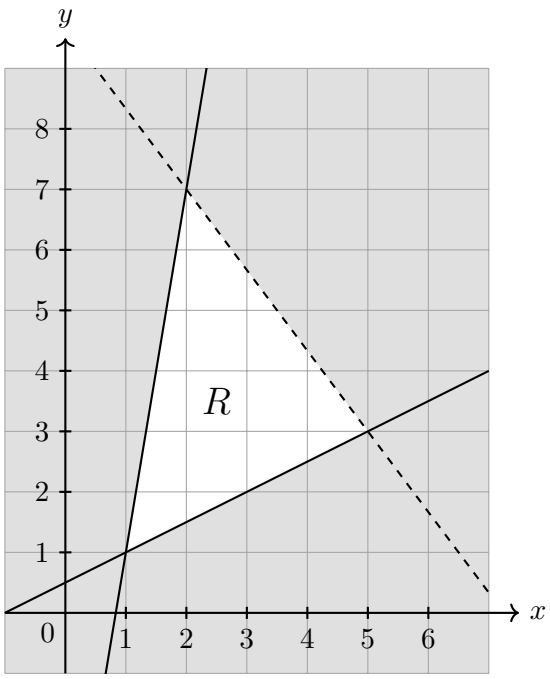
(i) Write down an inequality in terms of x and show that it simplifies to $x < 4.5$.

[2]

(ii) Given that x is a positive integer, find the maximum possible area of the garden.

Area = ... m^2 **[2]**

39 The diagram shows a coordinate grid.
The unshaded region R is bounded by three lines.



(a) Write down the three inequalities that define the region R.

.....

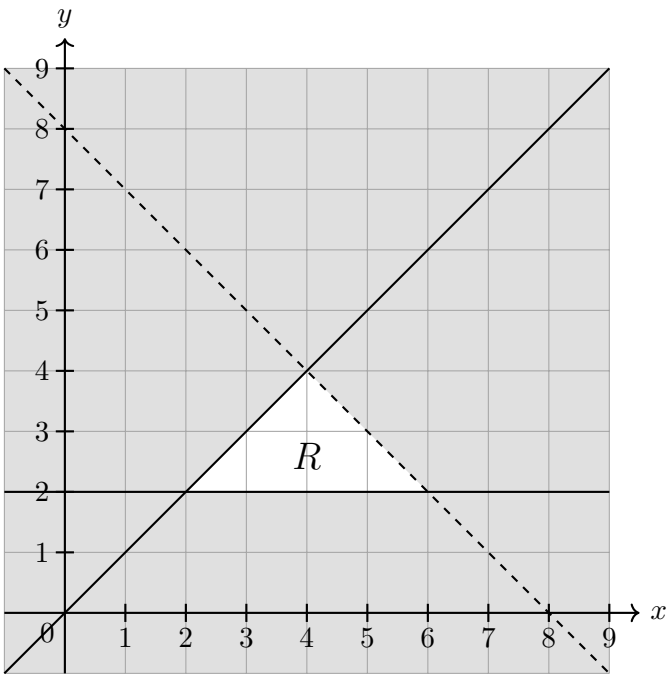
.....

..... **[3]**

(b) A point (x, y) in region R has integer coordinates.
Find the maximum possible value of $x + y$.

..... **[2]**

40 The diagram shows a grid with three lines drawn.
The unwanted regions are shaded grey.



(a) Write down the three inequalities that define the unshaded region R.

.....

.....

..... **[3]**

(b) Find the coordinates of the point (x, y) in region R where x and y are integers, such that the value of $2x + 3y$ is a maximum.

((..... , ...)) **[2]**

41 A factory produces two types of bicycles: Mountain bikes and City bikes.
In a week, the factory produces x Mountain bikes and y City bikes.

- The factory produces at least 15 Mountain bikes.
- The factory can produce at most 40 bicycles in a week.
- The number of City bikes produced is at least half the number of Mountain bikes produced.

(a) Write down three inequalities, in terms of x and y, to show this information.

.....

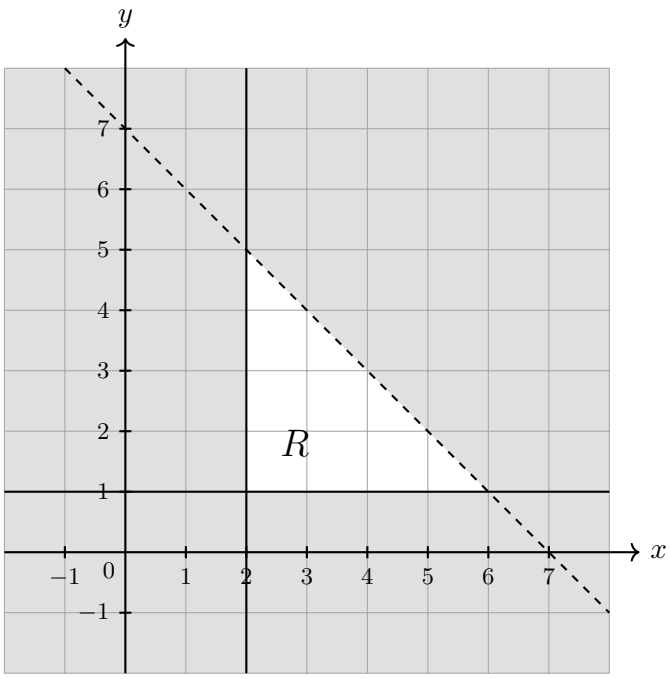
.....

..... **[3]**

(b) The profit on each Mountain bike is \$400 and the profit on each City bike is \$150.
Work out the maximum possible profit the factory can make in a week.

\$ ..... **[3]**

42 The unshaded region R in the diagram is defined by three inequalities.



(a) Write down the three inequalities that define the unshaded region R.

.....

.....

..... **[3]**

(b) Use a graphic display calculator to solve the inequality.

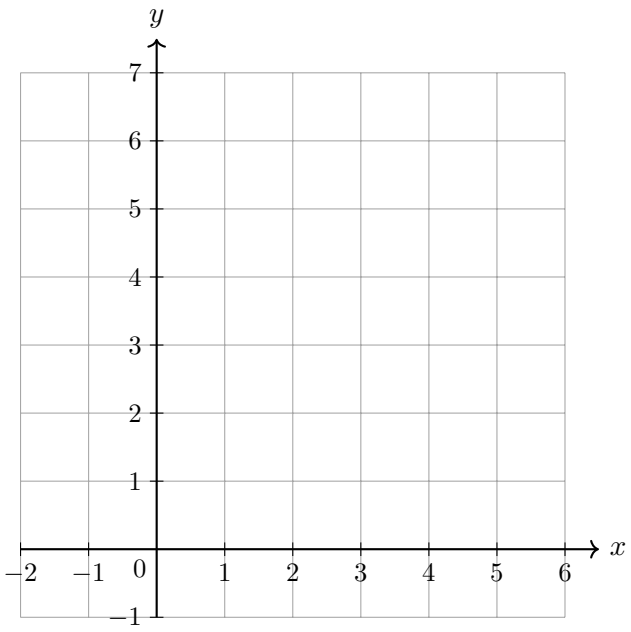
$x^2 - 4x - 2 \leq 0$

Give your answer correct to 2 decimal places.

..... **[3]**

43 **(a)** On the grid, draw the lines $y = 2$, $y = x + 2$, and $2x + y = 8$.
Shade the unwanted regions to show the region R defined by the inequalities:

- $y \geq 2$
- $y \leq x + 2$
- $2x + y \leq 8$



[4]

(b) Find the largest value of $3x + 2y$ for points (x, y) in the region R where both x and y are integers.

..... **[2]**

44 A rectangular garden has length $(2x + 5)$ metres and width $(x - 1)$ metres, where $x > 1$.
The perimeter of the garden is less than 50 metres and the area of the garden is at least 42 m^2.

(a) Use the information about the perimeter to show that $x < 7$.

[2]

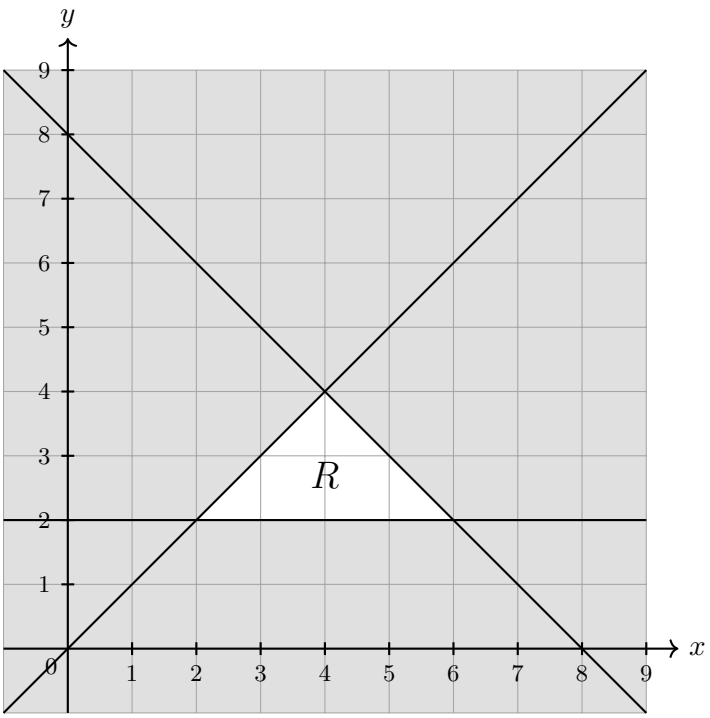
(b) Use the information about the area to show that $2x^2 + 3x - 47 \geq 0$.

[2]

(c) Use your graphic display calculator to solve the inequality $2x^2 + 3x - 47 \geq 0$ for $x > 1$, and hence state the range of possible values of x.
Give your boundary values correct to 2 decimal places where appropriate.

..... **[3]**

45 The diagram shows three straight lines on a coordinate grid.



(a) The region R is unshaded.
Write down the three inequalities that define the region R.

.....

.....

..... **[3]**

(b) Find the maximum value of $2x + 3y$ given that x and y are integers that satisfy these three inequalities.

..... **[3]**

(c) A point is chosen at random from the points with integer coordinates in region R.
Work out the probability that $x = y$.

..... **[2]**

46 **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

(b) Find the smallest integer x that satisfies the inequality.

$$\frac{2x-5}{3} - \frac{x-3}{4} > 1$$

$x =$ ..... **[3]**

(c) A taxi company, City Cabs, charges a fixed fare of \$3.20 plus \$1.50 per kilometre.
Another company, Metro Taxis, charges a fixed fare of \$5.00 plus \$1.10 per kilometre.
Construct and solve an inequality to find the minimum whole number of kilometres a passenger must travel for City Cabs to be more expensive than Metro Taxis.

..... kilometres **[3]**

47 Tickets for a community show cost \$5 for children and \$10 for adults.
The maximum number of tickets that can be sold is 500.
The organizers must sell at least 150 adult tickets.
The total amount of money raised from ticket sales must be at least \$3000.

Let x be the number of children's tickets sold and y be the number of adult tickets sold.

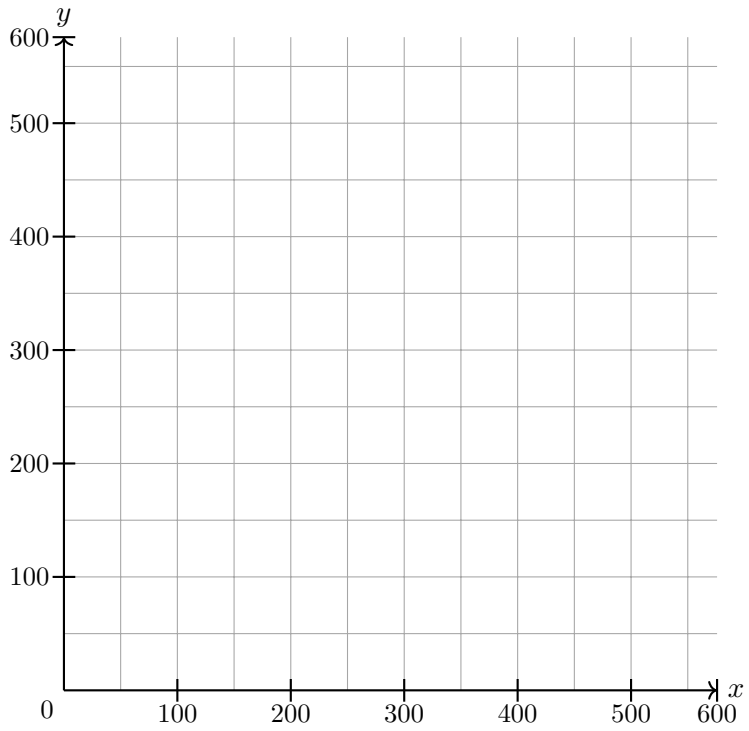
(a) Write down three inequalities in x and y to show this information.

.....

.....

..... **[3]**

(b) On the grid, draw three lines to show these inequalities.
Shade the unwanted regions to leave the region representing the possible number of tickets sold.



[4]

(c) The organizers want to maximize the number of children's tickets sold.
Find this maximum number of children's tickets and the corresponding number of adult tickets.

Children's tickets = ... , Adult tickets = ... **[2]**

48 A carpenter makes x tables and y chairs each week.
The total number of tables and chairs made each week must not exceed 60.
The number of tables made must be at least 10.
The number of chairs made must be at least half the number of tables made.

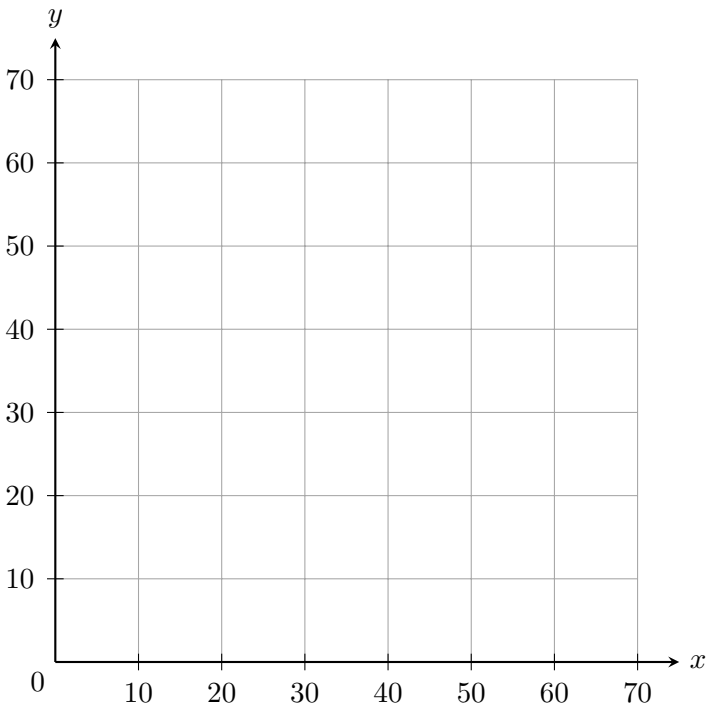
(a) Write down three inequalities, in terms of x and y, to show this information.

.....

.....

..... **[3]**

(b) On the grid, draw three straight lines and shade the unwanted regions to show the region which satisfies these inequalities.
Label the region R.



[4]

(c) The profit on each table is \$40 and the profit on each chair is \$25.
Work out the maximum possible weekly profit.

$ ..... **[2]**

49 A clothing company manufactures two types of shirts: Standard and Premium.
In one day, they make x Standard shirts and y Premium shirts.
They can make a maximum of 60 shirts in total per day.
Each Standard shirt requires 1 metre of fabric and each Premium shirt requires 2 metres of fabric.
The company has a maximum of 80 metres of fabric available per day.
They must make at least 10 Premium shirts per day.

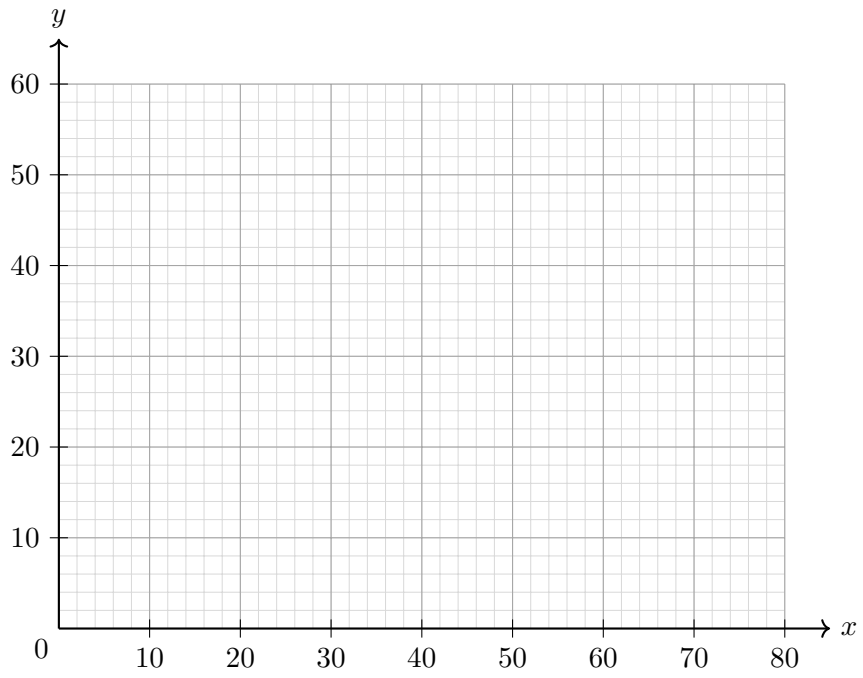
(a) Write down three inequalities in x and y, other than $x \geq 0$, to represent this information.

.....

.....

..... **[3]**

(b) On the grid provided, draw three lines to show these inequalities.
Shade the unwanted regions to leave the region that satisfies the inequalities unshaded.



[4]

(c) The profit on each Standard shirt is \$5 and the profit on each Premium shirt is \$8.
Find the maximum possible profit the company can make in one day.

$ ..... **[2]**

50 **(a)** Find the exact range of values of x for which

$$\frac{2x-3}{4} < \frac{x+5}{3} \leq \frac{3x-1}{2}$$

..... **[5]**

(b) A mobile phone company offers two plans.
Plan A costs \$12 per month plus \$0.07 per minute of calls.
Plan B costs \$25 per month for unlimited calls.

Let x be the number of minutes of calls in a month.

(i) Write down an inequality in terms of x to show the condition when Plan B is strictly cheaper than Plan A.

..... **[1]**

(ii) Solve your inequality to find the minimum number of whole minutes of calls required for Plan B to be cheaper than Plan A.

..... minutes **[3]**

51 **(a)** A factory makes x Mountain bicycles and y Road bicycles each day.

- The factory makes at most 60 bicycles in total each day.
- It makes at least 10 Road bicycles.
- It makes at least 15 Mountain bicycles.
- The number of Road bicycles is at most twice the number of Mountain bicycles.

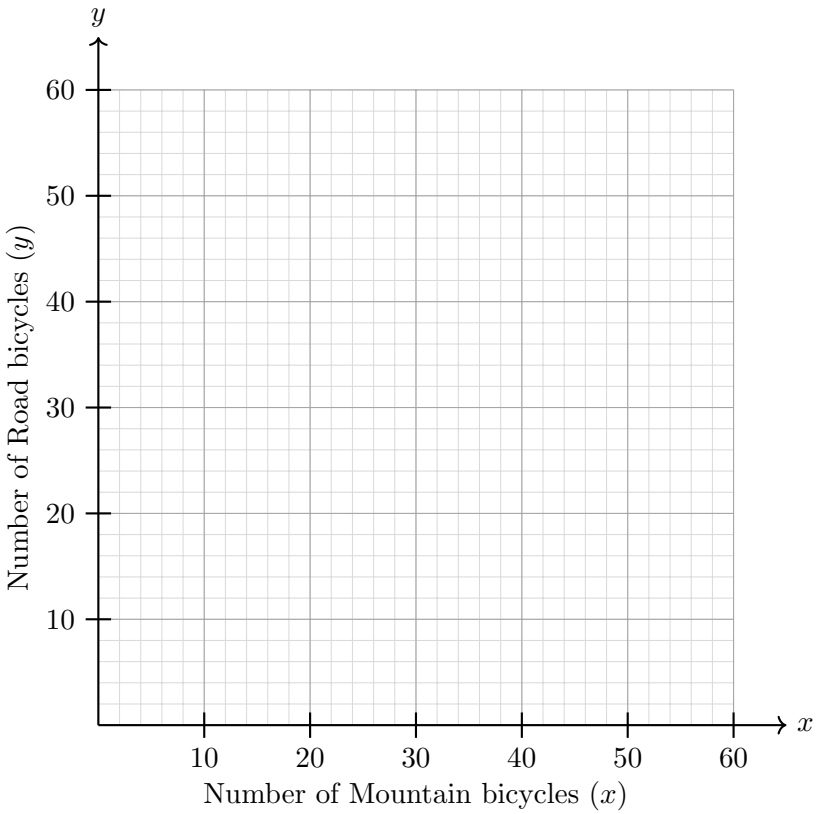
Write down four inequalities, in terms of x and y, to show this information.

.....

.....

..... **[4]**

(b) On the grid, draw four lines to show these inequalities.
By shading the unwanted regions, indicate the region R which satisfies all four inequalities.



[5]

(c) The profit on each Mountain bicycle is \$30 and the profit on each Road bicycle is \$50.
Find the maximum possible profit the factory can make in one day, and the number of each type of bicycle it must make to achieve this profit.

Maximum profit = \$ .....

Mountain bicycles = ...

Road bicycles = ... **[3]**

1 Rearranges to collect x terms on one side and constants on the other, e.g., $-8 < 4x$ or $-4x < 8$ **[M1]**

$x > -2$ or $-2 < x$ **[A1]**

[Back to Question] *[Total Marks: 2]*

2 (a) $3x - 6 > 5x + 4$ **[M1]**

$x < -5$ **[A1]**

(b) -6 **[B1]**

[Back to Question] *[Total Marks: 3]*

3 Expands brackets correctly: $12x - 4 - 5x \geq 2x + 16$ **[M1]**

Rearranges to collect x terms: $7x - 2x \geq 16 + 4$ or $5x \geq 20$ **[M1]**

$x \geq 4$ **[A1]**

[Back to Question] *[Total Marks: 3]*

4 (a) $3x - 12 < 5x - 2$

[M1]

$-10 < 2x \implies x > -5$ (or $-5 < x$)

[A1]

(b) -4

[B1]

[Back to Question] *[Total Marks: 3]*

5 (a) $6x - 3 > 4x + 7$ **[M1]**

$2x > 10$ leading to $x > 5$ **[A1]**

(b) Open circle placed at 5 with an arrow pointing to the right **[B1]**

(c) 6 **[B1]**

[Back to Question] *[Total Marks: 4]*

6 (a) $-6 < 3x \leq 12$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) $-1, 0, 1, 2, 3, 4$ **[B2]**

(or [B1] for 5 correct with no incorrect, or all 6 correct with one incorrect)

[Back to Question] *[Total Marks: 4]*

7 **(a)** $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 and closed circle at 4 **[B1]**

Thick line joining the two circles **[B1]**

[Back to Question] *[Total Marks: 4]*

8 **(a)** $4x \leq 20$ **[M1]**

$x \leq 5$ **[A1]**

(b) $3x + 6 < 20$ or $x + 2 < 6.66\ldots$ **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** $2x < 12$ **[M1]**

$x < 6$ **[A1]**

(b) Recognises open circle means strict inequality ($>$ or $<$) and arrow to right means greater than **[M1]**

$x > -2$ **[A1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** M1 for a correct first step, e.g., adding 1 to all parts to get $-4 < 2x \leq 8$

M1 for a correct second step, e.g., dividing all parts by 2

$-2 < x \leq 4$ **[A1]**

(b) M1 for an open circle at -2 and a closed solid circle at 4

A1 for a continuous thick line drawn between the two circles

[Back to Question] *[Total Marks: 5]*

11 **(a)** Expands brackets correctly: $3x - 6 \leq 5x + 8$ **[M1]**

Rearranges to isolate x, e.g. $-14 \leq 2x$ or $-2x \leq 14$ **[M1]**

$x \geq -7$ **[A1]**

(b) Places a closed, solid circle at -7 **[B1]**

Draws a line with an arrow pointing to the right from the circle **[B1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** Adds 1 to all three parts: $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 and closed (filled) circle at 4 **[B1]**

Horizontal line segment connecting the circles at -2 and 4 **[B1]**

(c) -1 **[B1]**

[Back to Question] *[Total Marks: 5]*

13 (a) $-9 > 2x$ or $-2x > 9$ **[M1]**

$x < -4.5$ **[A1]**

(b) $-4 < 2n \leq 4$ **[M1]**

$-2 < n \leq 2$ **[M1]**

$-1, 0, 1, 2$ **[A1]**

(c) $x > -1$ (accept any valid variable) **[B1]**

[Back to Question] *[Total Marks: 6]*

14 (a) $-15 \leq 3x$ or $-3x \leq 15$ **[M1]**

$x \geq -5$ or $-5 \leq x$ **[A1]**

(b) $-7 < 2x \leq 8$ **[M1]**

$-3.5 < x \leq 4$ **[A1]**

(c) Largest $= 4$ **[B1]**

Smallest $= -3$ **[B1]**

[Back to Question] *[Total Marks: 6]*

15 (a) $-3x < 9$ or $2 - 11 < 3x$ **[M1]**

$x > -3$ **[A1]**

(b) $8x - 4 \leq 5x + 17$ **[M1]**

$3x \leq 21$ **[M1]**

$x \leq 7$ **[A1]**

(c) $-3 < n \leq 7$ **[M1]**

$-2, -1, 0, 1, 2, 3, 4, 5, 6, 7$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 (a) Subtracts $3x$ and 3 to reach $\pm 4x \leq \pm 8$ or similar **[M1]**

$x \geq -2$ **[A1]**

(b) Solid circle (or equivalent clear inclusive mark) placed at -2 **[B1]**

Thick line or arrow pointing to the right starting from -2 **[B1]**

(c) Adds 5 to all parts to get $3 < 3x \leq 9$ **[M1]**

Divides by 3 to get $1 < x \leq 3$ **[A1]**

$2, 3$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 $y \geq 1$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 6$ **[B1]**

[Back to Question] *[Total Marks: 3]*

18 $y \leq 5x - 4$ or equivalent **[B1]**

$2y \geq x + 1$ or equivalent **[B1]**

$x + y \leq 8$ or equivalent **[B1]**

[Back to Question] *[Total Marks: 3]*

19 $x \geq 2$ **[B1]**

$y > x$ **[B1]**

$x + y \leq 8$ (or $y \leq -x + 8$) **[B1]**

[Back to Question] *[Total Marks: 3]*

20 $x \geq 1$

[B1]

$y \geq 2$

[B1]

$x + y < 6$ (or $y < -x + 6$)

[B1]

[Back to Question] *[Total Marks: 3]*

21 **(a)** $-6 < 3x \leq 12$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 **[B1]**

Closed circle at 4 and a line joining the two circles **[B1]**

[Back to Question] *[Total Marks: 4]*

22 **(a)** $-6 < 2x \leq 6$ (Allow M1 for one correct side e.g. $-6 < 2x$ or $2x \leq 6$) **[M2]**

$-3 < x \leq 3$ **[A1]**

(b) Open circle at -3 and closed circle at 3 **[B1]**

A single continuous line drawn connecting the two circles **[B1]**

[Back to Question] *[Total Marks: 5]*

23 **(a)** **(a)** Adds 1 to all three parts: $-4 \leq 2x < 7$ **[M1]**

Divides all three parts by 2 **[M1]**

$-2 \leq x < 3.5$ or $-2 \leq x < \frac{7}{2}$ **[A1]**

(b) Award [B1] for at least three correct integers and no incorrect integers, or all correct integers with only one incorrect integer

$-2, -1, 0, 1, 2, 3$ **[B2]**

[Back to Question] *[Total Marks: 5]*

24 **(a)** $x < 4$ **[B1]**

$y \geq -x + 3$ **[B1]**

$y \leq \frac{1}{2}x + 3$ (or equivalent, e.g. $2y \leq x + 6$) **[B1]**

(b) Substitutes $x = 3$ into the inequalities: $b \geq -3 + 3 = 0$ and $b \leq 0.5(3) + 3 = 4.5$ **[M1]**

Deduces the range $0 \leq b \leq 4.5$ **[A1]**

$0, 1, 2, 3, 4$ **[A1]**

[Back to Question] *[Total Marks: 6]*

25 **(a)** $y \geq 0$ **[B1]**

Equation of sloping boundary identified as $x + y = 6$ or $y = -x + 6$ **[B1]**

$x + y < 6$ or $y < -x + 6$ (correct strict inequality symbol) **[B2]**

(b) Identifies valid integer coordinate $(1, 4)$ (or tests boundary lattice points such as $(1, 4), (2, 3), (3, 2)$) **[M1]**

14 **[A1]**

[Back to Question] *[Total Marks: 6]*

26 $x \geq 1$ **[B1]**

$y \geq 2$ **[B1]**

$x + y < 6$ or $y < -x + 6$ **[B1]**

[Back to Question] *[Total Marks: 3]*

27 **(a)** $y \geq 1$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 6$ or $y \leq -x + 6$ **[B1]**

(b) Tests a valid integer coordinate close to the boundary, e.g., $(5, 1)$, $(4, 2)$, or $(3, 3)$ **[M1]**

17 (from $x = 5, y = 1$) **[A1]**

[Back to Question] *[Total Marks: 5]*

28 Identifies horizontal line $y = 2$ **[M1]**

States $y \geq 2$ (accept $y > 2$) **[A1]**

Identifies dashed line passing through $(0, 6)$ and $(4, 2)$ **[M1]**

Finds equation $y = -x + 6$ and states $y < -x + 6$ **[A1]**

Identifies solid line passing through $(0, 2)$ and $(1, 5)$ **[M1]**

Finds equation $y = 3x + 2$ and states $y \leq 3x + 2$ **[A1]**

[Back to Question] *[Total Marks: 6]*

29 **(a)** Multiplies by 15 to clear denominators: $3(3x - 2) - 5(2x - 5) \leq 30$ **[M1]**

$9x - 6 - 10x + 25 \leq 30$ **[M1]**

$-x + 19 \leq 30$ **[M1]**

$x \geq -11$ **[A1]**

(b) Identifies that y can only take integer values of 2 or 3 (since $2y > x + y > 2 \implies y > 1$) **[M1]**

Finds correct points for one of the valid y values: e.g. $(1, 2)$ for $y = 2$ OR $(0, 3), (1, 3), (2, 3)$ for $y = 3$ **[B1]**

States all four correct points: $(1, 2), (0, 3), (1, 3), (2, 3)$ with no extras **[A1]**

[Back to Question] *[Total Marks: 7]*

30 **(a)** Rearranging terms to get $3x \leq 15$ **[M1]**

$x \leq 5$ **[A1]**

(b) $1, 2, 3, 4, 5$ **[B2]**

If 0 scored, award 1 mark for $0, 1, 2, 3, 4, 5$ OR for omitting exactly one correct value **[B1]**

[Back to Question] *[Total Marks: 4]*

31 $y \geq 2$ **[B1]**

$y \leq 2x$ **[B1]**

$x + y \leq 6$ **[B1]**

[Back to Question] *[Total Marks: 3]*

32 $x \geq 1$ **[B1]**

$y \leq 2x$ **[B1]**

$x + y \leq 6$ **[B1]**

(Note: Accept strict inequalities $<, >$)

[Back to Question] *[Total Marks: 3]*

33 (a) Subtracting 2 from all parts: $-9 < 3x \leq 9$ **[M1]**

Dividing by 3: $-3 < x$ and $x \leq 3$ **[M1]**

$-3 < x \leq 3$ **[A1]**

(b) 3 **[B1]**

[Back to Question] *[Total Marks: 4]*

34 (a) Adding 1 to all parts: $-6 \leq 3x < 12$ **[M2, M1 for one side correct]**

Dividing by 3: $-2 \leq x < 4$ **[A1]**

(b) Identifies at least 3 correct integers or includes 4 but excludes -2 **[M1]**

$-2, -1, 0, 1, 2, 3$ **[A1]**

[Back to Question] *[Total Marks: 5]*

35 (a) $y \geq 1$ **[B1]**

$y \leq x + 1$ **[B1]**

$y < 5 - x$ (or $x + y < 5$) **[B1]**

All three inequality signs correct (with strict inequality for the dashed line) **[B1]**

(b) Identifies $(1, 2)$ as a valid integer coordinate point inside R **[M1]**

$(1, 2)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

36 (a) Subtracts 1 from all parts: $-6 < 2x \leq 10$ **[M1]**

Divides by 2 across all parts **[M1]**

$-3 < x \leq 5$ **[A1]**

(b) Identifies at least two correct integers from the set **[M1]**

$-2, -1, 0$ **[A1]**

(c) Finds the correct roots of the quadratic $2x^2 - 7x + 3 = 0$ as 0.5 and 3 **[B1]**

$x < 0.5$ or $x > 3$ **[B1]**

[Back to Question] *[Total Marks: 7]*

37 **(a)** $y \geq 1$ **[B1]**

$y \leq 2x - 2$ **[B2]**

(or $M1$ for finding the boundary equation $y = 2x - 2$)

$x + y \leq 7$ (or $y \leq -x + 7$) **[B2]**

(or $M1$ for finding the boundary equation $x + y = 7$)

(b) Identifies a valid integer coordinate inside or on the boundary of R that maximizes the expression, e.g., $(6, 1)$, $(5, 2)$, or $(4, 3)$ **[M1]**

13 (from $2(6) + 1 = 13$) **[A1]**

[Back to Question] *[Total Marks: 7]*

38 **(a)** $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) $-1, 0, 1, 2, 3, 4$ **[B2]**

1 omission or 1 extra integer **[B1]**

(c) **(i)** $2(3x + 2) + 2x < 40$ or $(3x + 2) + x < 20$ **[M1]**

$8x + 4 < 40$

8x ¡ 36

x ¡ $4.5 with outer errors [A1]$

(ii) Identifies maximum integer $x = 4$ **[M1]**

Length = 14, Width = 4, Area = 56 **[A1]**

[Back to Question] *[Total Marks: 8]*

39 **(a)** $y \geq 0.5x + 0.5$ or $2y \geq x + 1$ (Accept equivalent solid boundary) **[B1]**

$y \leq 6x - 5$ (Accept equivalent solid boundary) **[B1]**

$y < -\frac{4}{3}x + \frac{29}{3}$ or $4x + 3y < 29$ (Must be strict inequality for dashed line) **[B1]**

(b) Identifies a valid integer point near the boundary that maximizes the sum, e.g., $(2, 6), (3, 5)$, or $(4, 4)$. Note: $(2, 7)$ is invalid as it lies on the dashed boundary **[M1]**

8 **[A1]**

[Back to Question] *[Total Marks: 5]*

40 **(a)** $y \geq 2$ **[B1]**

$y \leq x$ **[B1]**

$x + y < 8$ **[B1]**

(b) Identifies $(4, 3)$ or $(5, 2)$ as integer coordinates within the region R. Note that $(4, 4)$ is excluded due to the strict inequality $x + y < 8$. **[M1]**

Evaluates $2x + 3y$ and correctly identifies $(4, 3)$ as yielding the maximum value (17). **[A1]**

[Back to Question] *[Total Marks: 5]*

41 **(a)** $x \geq 15$ **[B1]**

$x + y \leq 40$ **[B1]**

$y \geq \frac{1}{2}x$ or $2y \geq x$ **[B1]**

(b) Identifies the objective function for profit: $P = 400x + 150y$ **[M1]**

Tests the valid integer vertex $(26, 14)$ or evaluates the profit equation showing profit is maximised at the highest possible integer value for x **[M1]**

12500 **[A1]**

[Back to Question] *[Total Marks: 6]*

42 **(a)** $x \geq 2$ **[B1]**

$y \geq 1$ **[B1]**

$x + y < 7$ or $y < -x + 7$ **[B1]**

(b) Finding critical values -0.45 and 4.45 (allow ≈ -0.449 and ≈ 4.449) **[B2]**

(Allow B1 for setting $x^2 - 4x - 2 = 0$ and finding one correct root or correct quadratic formula setup)

$-0.45 \leq x \leq 4.45$ **[B1]**

[Back to Question] *[Total Marks: 6]*

43 **(a)** Draw correct solid line $y = 2$

[B1]

Draw correct solid line $y = x + 2$

[B1]

Draw correct solid line $2x + y = 8$

[B1]

Correct shading of unwanted regions, leaving the triangle with vertices $(0, 2)$, $(3, 2)$, and $(2, 4)$ unshaded and clearly representing R.

[B1]

(b) Identify integer coordinates $(2, 4)$ or $(3, 2)$ as yielding the maximum value

[M1]

14

[A1]

[Back to Question] *[Total Marks: 6]*

44 **(a)** $2[(2x + 5) + (x - 1)] < 50$ or $6x + 8 < 50$ **[M1]**

$6x < 42 \implies x < 7$ **[A1]**

(b) $(2x+5)(x-1) \geq 42$ **[M1]**

$2x^2 + 3x - 5 \geq 42 \implies 2x^2 + 3x - 47 \geq 0$ **[A1]**

(c) Finds positive root of $2x^2 + 3x - 47 = 0$ as 4.155... (4.16) **[M1]**

Solves quadratic inequality for $x > 1$: $x \geq 4.16$ **[A1]**

$4.16 \leq x < 7$ **[A1]**

[Back to Question] *[Total Marks: 7]*

45 (a) $y \geq 2$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 8$ **[B1]**

(b) Identifies at least two valid integer coordinate pairs within the region R (e.g., $(6,2)$, $(5,3)$, $(4,4)$) **[M1]**

Evaluates $2x + 3y$ for at least one critical integer point, e.g., $2(4) + 3(4) = 20$ or $2(5) + 3(3) = 19$ **[M1]**

20 **[A1]**

(c) Identifies the correct total number of integer points in R is 9, or identifies the 3 correct points where $x = y$: $(2,2), (3,3), (4,4)$ **[M1]**

$\frac{3}{9}$ or $\frac{1}{3}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

46 (a) $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Multiplies throughout by 12 (or uses common denominator): $4(2x-5) - 3(x-3) > 12$ **[M1]**

Expands and simplifies: $8x - 20 - 3x + 9 > 12 \implies 5x - 11 > 12 \implies 5x > 23$ **[M1]**

5 **[A1]**

(c) Constructs correct inequality: $3.20 + 1.50k > 5.00 + 1.10k$ **[M1]**

Solves for k: $0.40k > 1.80 \implies k > 4.5$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 8]*

47 (a) $x + y \leq 500$ **[B1]**

$y \geq 150$ **[B1]**

$5x + 10y \geq 3000$ (or simplified to $x + 2y \geq 600$) **[B1]**

(b) Line drawn correctly for $x + y = 500$ (passing through $(0, 500)$ and $(500, 0)$) **[B1]**

Line drawn correctly for $y = 150$ (horizontal line) **[B1]**

Line drawn correctly for $x + 2y = 600$ (passing through $(0, 300)$ and $(600, 0)$) **[B1]**

Correct region identified and all unwanted regions properly shaded (leaving quadrilateral with vertices $(0, 300), (0, 500), (350, 150), (300, 150)$ unshaded) **[B1]**

(c) Identifies the vertex $(350, 150)$ as giving the maximum x value **[M1]**

Children's tickets = 350, Adult tickets = 150 **[A1]**

[Back to Question] *[Total Marks: 9]*

48 **(a)** $x + y \leq 60$ **[B1]**

$x \geq 10$ **[B1]**

$y \geq 0.5x$ (or $2y \geq x$ or $y \geq \frac{1}{2}x$) **[B1]**

(b) Draws the line $x + y = 60$ correctly (from $(0, 60)$ to $(60, 0)$) **[B1]**

Draws the line $x = 10$ correctly (vertical line at 10) **[B1]**

Draws the line $y = 0.5x$ correctly (from $(0, 0)$ passing through $(20, 10)$, $(40, 20)$, etc.) **[B1]**

Shades the unwanted regions correctly, leaving the correct triangular region R unshaded **[B1]**

(c) Identifies $(40, 20)$ as the vertex for maximum profit (or tests multiple vertices) **[M1]**

2100 **[A1]**

[Back to Question] *[Total Marks: 9]*

49 **(a)** $x + y \leq 60$ **[B1]**

$x + 2y \leq 80$ **[B1]**

$y \geq 10$ **[B1]**

(b) B1 for correctly drawing the line $x + y = 60$

B1 for correctly drawing the line $x + 2y = 80$

B1 for correctly drawing the line $y = 10$

B1 for correctly shading the unwanted regions (above the two diagonal lines, and below the horizontal line) to leave the correct valid region unshaded

(c) M1 for a valid method to find the maximum profit, e.g., testing the vertex $(40, 20)$ in the objective function $5x + 8y$

360 **[A1]**

[Back to Question] *[Total Marks: 9]*

50 **(a)** Forms left inequality: $3(2x - 3) < 4(x + 5)$ or equivalent **[M1]**

Simplifies to $6x - 9 < 4x + 20$

x ¡ $\frac{29}{2}$ (or 14.5) [A1]

Forms right inequality: $2(x+5) \leq 3(3x-1)$ or equivalent **[M1]**

Simplifies to $2x + 10 \leq 9x - 3$

x $\geq \frac{13}{7}$ [A1]

$\frac{13}{7} \leq x < \frac{29}{2}$ (accept $1.86 \leq x < 14.5$ or equivalent double inequality) **[A1]**

(b) (i) $25 < 12 + 0.07x$ (or $12 + 0.07x > 25$) **[B1]**

(ii) $13 < 0.07x$ **[M1]**

$x > 185.71\ldots$ **[A1]**

186 **[A1]**

[Back to Question] *[Total Marks: 9]*

51 (a) $x + y \leq 60$ **[B1]**

$y \geq 10$ **[B1]**

$x \geq 15$ **[B1]**

$y \leq 2x$ **[B1]**

(b) Line $x + y = 60$ drawn correctly **[B1]**

Line $y = 10$ drawn correctly **[B1]**

Line $x = 15$ drawn correctly **[B1]**

Line $y = 2x$ drawn correctly **[B1]**

Correct region R indicated with all four unwanted regions shaded **[B1]**

(c) Identifies $(20, 40)$ as the vertex for maximum profit **[M1]**

Mountain bicycles = 20 and Road bicycles = 40 **[A1]**

Maximum profit = 2600 **[A1]**

[Back to Question] *[Total Marks: 12]*